Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Official Title: A Randomized Phase II, Double-blind, Multicenter Study Evaluating the Efficacy and Safety of Autogene Cevumeran Plus Nivolumab Versus Nivolumab as Adjuvant Therapy in Patients With High-risk Muscle-invasive Urothelial Carcinoma **Brief Title:** A Study to Evaluate the Efficacy and Safety of Autogene Cevumeran With Nivolumab Versus Nivolumab Alone in Participants With High-Risk Muscle-Invasive Urothelial Carcinoma (MIUC) **Short Title/ Acronym:** IMCODE004 **Protocol Number:** BO45230 **NCT Number:** NCT06534983 **Sponsor/Company:** Roche (Hoffmann-La Roche) **Investigational Product:** Autogene Cevumeran (RO7198457) + Nivolumab **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy of adjuvant treatment with the personalized mRNA cancer vaccine autogene cevumeran plus nivolumab compared with nivolumab alone on investigator-assessed disease-free survival (DFS). * **Secondary Objectives:** To evaluate Overall Survival (OS), Investigator Assessed DFS in PD-L1 Expression $\ge$ 1% Population, Investigator Assessed Distant Metastasis-Free Survival (DMFS), and the overall safety/tolerability of the combination therapy.

2.2 Background & Rationale To address the high rate of disease recurrence in patients with high-risk Muscle-Invasive Urothelial Carcinoma (MIUC) following surgical resection. While adjuvant nivolumab improves disease-free survival, a significant unmet need remains for patients with high-risk features. This study investigates whether the addition of autogene cevumeran, an individualized neoantigen specific immunotherapy (mRNA vaccine), can synergize with immune checkpoint blockade (nivolumab) to further delay or prevent recurrence and extend survival.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Saline/Sodium Chloride + Nivolumab) * **Blinding:** Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor) / Double-blind * **Randomization:** Randomized (Parallel Assignment) * **Trial Status:** Recruiting (Currently recruiting participants)

3.2 Planned Enrollment & Duration * **Targeted Enrollment:** 362 participants * **Number of Sites:** Multicenter (Global) * **Estimated Study Start:** 09-Dec-2024 * **Estimated Primary Completion:** 07-Mar-2029

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must have the capacity to participate/enroll in the study and to provide informed consent. 2. Histologically confirmed muscle-invasive UC (also termed TCC) of the bladder or upper urinary tract. 3. TNM classification (UICC/AJCC 7th edition) at pathological examination of surgical resection specimen of (y)pT3-4 or (y)pN+ and M0. 4. Surgical resection of MIUC of the bladder or upper tract. 5. Participants who have not received prior neoadjuvant cisplatin chemotherapy (NAC) must be ineligible to receive adjuvant cisplatin therapy due to participant refusal, cisplatin ineligibility or investigator decision. 6. Tumor tissue must be provided for biomarker analysis. 7. Absence of residual disease and absence of metastasis, as confirmed by a negative baseline Computed Tomography (CT) or Magnetic Resonance Imaging (MRI) scan of the pelvis, abdomen, and chest no more than 28 days prior to randomization. 8. Full recovery from cystectomy or nephroureterectomy within 120 days following surgery. 9. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 10. Negative HIV test at screening. 11. Negative hepatitis B surface antigen (HbsAg) test at screening. 12. Positive hepatitis B surface antibody (HBsAb), or a negative HBsAb at screening accompanied by either of the following: negative total hepatitis B core antibody (HBcAb) or positive total HBcAb test followed by quantitative hepatitis B virus (HBV) DNA < 500 international units/milliliter (IU/mL). 13. Negative hepatitis C virus (HCV) antibody test at screening, or a positive HCV antibody test followed by a negative HCV RNA test at screening.

4.2 Exclusion Criteria 1. Partial cystectomy in the setting of bladder cancer primary tumor or partial nephroureterectomy in the setting of renal pelvis primary tumor. 2. Any approved anti-cancer therapy, including chemotherapy, or hormonal therapy within 3 weeks prior to initiation of study treatment. 3. Any prior neoadjuvant immunotherapy. 4. Adjuvant chemotherapy or radiation therapy for UC following surgical resection. 5. Malignancies other than UC within 5 years prior to randomization. 6. Treatment with a live, attenuated vaccine within 4 weeks prior to initiation of study treatment.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Experimental Arm:* Autogene cevumeran administered at specified timepoints + Nivolumab (480 mg) once every 4 weeks (Q4W). * *Active Comparator*

Arm: Saline (Sodium Chloride) solution administered at specified timepoints + Nivolumab (480 mg) once every 4 weeks (Q4W). * **Route of Administration:** Intravenous (IV) infusion for both medications. * **Duration of Treatment:** 1 year of adjuvant therapy (or until disease recurrence or unacceptable toxicity).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care medications following cystectomy/nephroureterectomy. * **Prohibited:** Live attenuated vaccines, concurrent systemic immunosuppressive therapy, and other adjuvant chemotherapy/radiation.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, Baseline Imaging (CT/MRI), HIV/HBV/HCV testing, Pathology confirmation
Baseline	Day 0	Randomization, First Dose (Safety run-in phase initially, then randomized phase)
Treatment	Q4W	IV Infusions (Vaccine/Saline + Nivolumab), Safety Labs, Efficacy Assessment (Scans)
End of Treatment	1 Year	Final Treatment Visit, Transition to long-term follow up
Follow-Up	Post-Treatment	Ongoing imaging and survival tracking until disease recurrence or death

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Investigator Assessed Disease Free Survival (INV-DFS):** Time from randomization until the first recurrence of disease or death from any cause. Disease recurrence is defined as any of the following: * Local (pelvic) recurrence of urothelial carcinoma (UC) (including soft tissue and regional lymph nodes) * Urinary tract recurrence of UC (excluding low-grade non-muscle-invasive bladder cancer (NMIBC)) * Distant metastasis of UC

7.2 Secondary & Exploratory Endpoints 1. **Overall Survival (OS)** 2. **Investigator (INV) Assessed DFS in Programmed Death Ligand-1 (PD-L1) Expression $\ge$ 1% Population** 3. **Investigator (INV) Assessed Distant Metastasis-Free Survival (DMFS)**

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of all adverse events with a specific focus on immune-related adverse events (irAEs) and vaccine-related toxicities. Includes an initial safety run-in phase to ensure tolerability before opening randomization.
- **Serious Adverse Events (SAEs):** Standard reporting timelines strictly enforced per GCP guidelines.

- **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to oversee the safety run-in and ongoing blinded trial data.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy (DFS); Safety population for all participants receiving at least one dose.
- **Sample Size Justification:** Target $N = 362$ designed to provide sufficient statistical power to detect a significant hazard ratio (HR) improvement in DFS for the combination arm versus the nivolumab monotherapy arm.
- **Handling of Missing Data:** Censoring applied for DFS/OS per standard Kaplan-Meier methodology at the date of the last evaluable tumor assessment.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine rigorous onsite and remote clinical monitoring by the sponsor (Hoffmann-La Roche).
- **Audit Trail:** Validated eCRF with full audit trails, strict source data verification (SDV), and independent radiological review protocols.

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11. Study Identifiers & Governance

- **Primary ID (Protocol):** BO45230
- **Registry Number (NCT):** NCT06534983
- **Posting ID:** 475504151196251922
- **Sponsor/Lead Organization:** Roche
- **Study Title (Acronym):** IMCODE004

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12. Clinical Framework (Oncology/MIUC Specific)

- **Primary Condition/Disease Area:** Muscle Invasive Urothelial Carcinoma (Preferred UMLS: Muscle-Invasive Urothelial Carcinoma)
- **Diagnosis Threshold:** Post-surgical resection (y)pT3-4 or (y)pN+ and M0 status
- **Medical Methodology:** Adjuvant immunotherapy combined with individualized neoantigen specific immunotherapy
- **Optimization Target:** Eradication of micrometastatic disease and prevention of clinical recurrence

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13. Study Procedures & Methodology

- **Management Pathway:** Post-cystectomy/nephroureterectomy adjuvant care
- **Education Protocol (HCP):** Site initiation visits covering study drug preparation, IV infusion safety, and AE management.
- **Education Protocol (Patient):** Rigorous informed consent focusing on the experimental nature of personalized mRNA vaccines and immune checkpoint inhibitors.
- **Quality Audit Mechanism:** Centralized RECIST/DFS tracking and independent safety evaluations.

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14. Observation & Follow-Up Schedule

- **Total Duration:** Approximately 6 years (Randomization until the first recurrence of disease or death)
- **Onsite Visit Cadence:** Every 4 weeks (Q4W) during the 1-year treatment phase
- **Remote Monitoring Frequency:** Routine survival/status calls following the treatment period.
- **Checklist Review Interval:** Regular imaging intervals as defined by protocol to assess DFS.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]